

RNA->Protein (Translation and the Genetic Code)

Introduction

Translation is often the last step in the expression of genetic information. The sequence of nucleotides of the mRNA is read by a special translating machine, the ribosome, and used to assemble the linear sequence of amino-acids that make up a protein. The correspondence between the linear nucleotide sequence and the linear amino-acid sequence is called the genetic code. As you study this process, try working in stages. In the first stage, focus on the information transfer from linear string of nucleotides to amino acids, without worrying about the how. Then focus on the skeleton of the mechanism, the interactions between the mRNAs and tRNAs and between the tRNAs and the amino acids. Finally, add in the machinery that enables these, the aminoacyl tRNA synthetases and the ribosome. Follow the information!

Reading

MBoC(6th) Ch6: FROM RNA TO PROTEIN. There is more material covered in the chapters than you need to know. Use the text and illustrations below to help focus.

Need to know and understand:

RNA template structure

mRNAs

Ribosome Binding Site (RBS): Shine-Delgarno sequence in E.coli, Kozak sequence in Eukaryotes

5' and 3' untranslated regions (UTR)

5' AUG initiation codon

open reading frame (ORF)

coding sequence

UAA, UAG, UGA stop codons

Cistron, Polycistron

Ribosome translation machinery

Ribosome - different subunits. Made of ribosomal RNAs (rRNA) + proteins

Entry of Ribosomes onto mRNA. Prokaryote ribosomes can bind directly to RBS.

Eukaryote ribosomes bind to 5' mRNA cap and slide to first RBS.

Ribosomes complete assembly only after binding to mRNA.

Initiation at AUG start codon requires initiation factors IF1, IF2, IF3 to be present.

mRNA is read from 5' to 3'.

tRNAs move from A->P->E sites in ribosome during translation

peptidyl transferase activity (peptide bond catalysis) by a rRNA (23S rRNA in *E. coli*). It's a ribozyme!

Peptide release factors RF₁, RF₂, RF₃

tRNAs

tRNA adapter hypothesis

codon loop

acceptor end

There are fewer tRNAs than codons. Some tRNAs contain modified nucleotides, eg.

inosine that are able to base pair with more than one base on the mRNA. These are usually found in the wobble position. The result is that one tRNA may recognize several codons that all code for the same amino acid.

Charged amino-acyl tRNAs. amino-acyl bond between 3' OH of A at 3' end of tRNA and amino group of amino acid is high energy, higher than the peptide bond between amino acids. Charging is specific between tRNAs and amino acids catalysed by amino-acyl tRNA synthetases. These enzymes specifically recognize both the tRNAs and the amino

acid and attaches the amino acid to the 3' A that is found in all tRNAs. Often the same synthetase will recognize and charge several tRNAs that recognize different codons, but code for the same amino acid (isoacceptors).

Genetic code

Universality (except in mitochondria and a few others)

Codon

Anticodon

reading frame

Wobble position

degeneracy

AUG initiation codon = start codon (not always true)

TGA, TAG, TAA termination codon = stop codons

Differences in translation between eukaryotes and Prokaryotes

(Medical Perspectives) More than either the machinery of replication or transcription, the ribosome is a rich target for antibiotics that preferentially block prokaryotic translation. Tetracycline, Streptomycin, Chloramphenicol, and Erythromycin are all classic antibiotics that do so. The initiation of translation is also coupled to innate immune system activation. A virus-infected host cell often shuts down the start of translation by inactivating one of the eIFs (See fig 6-70) through phosphorylation. Finally, the Diphtheria bacillus secretes a toxin that inhibits EF2, preventing the ribosome from advancing during translation.

Focus illustrations: Source: Molecular Biology of the Cell: Sixth Edition (2015) Alberts et al., Garland Science, NY

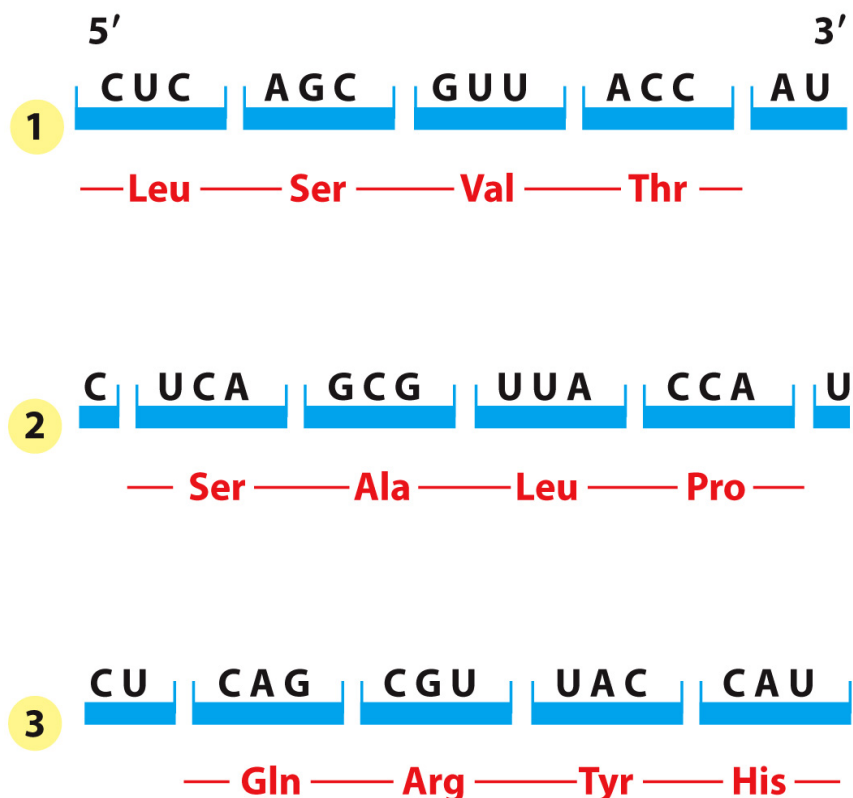


Figure 6-49 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 6-49 The three possible reading frames in protein synthesis.

In the process of translating a nucleotide sequence (blue) into an amino acid sequence (red), the sequence of nucleotides in an mRNA molecule is read from the 5' end to the 3' end in consecutive sets of three nucleotides. In principle, therefore, the same RNA sequence can specify three completely different amino acid sequences, depending on the reading frame. In reality,

however, only one of these reading frames contains the actual message.

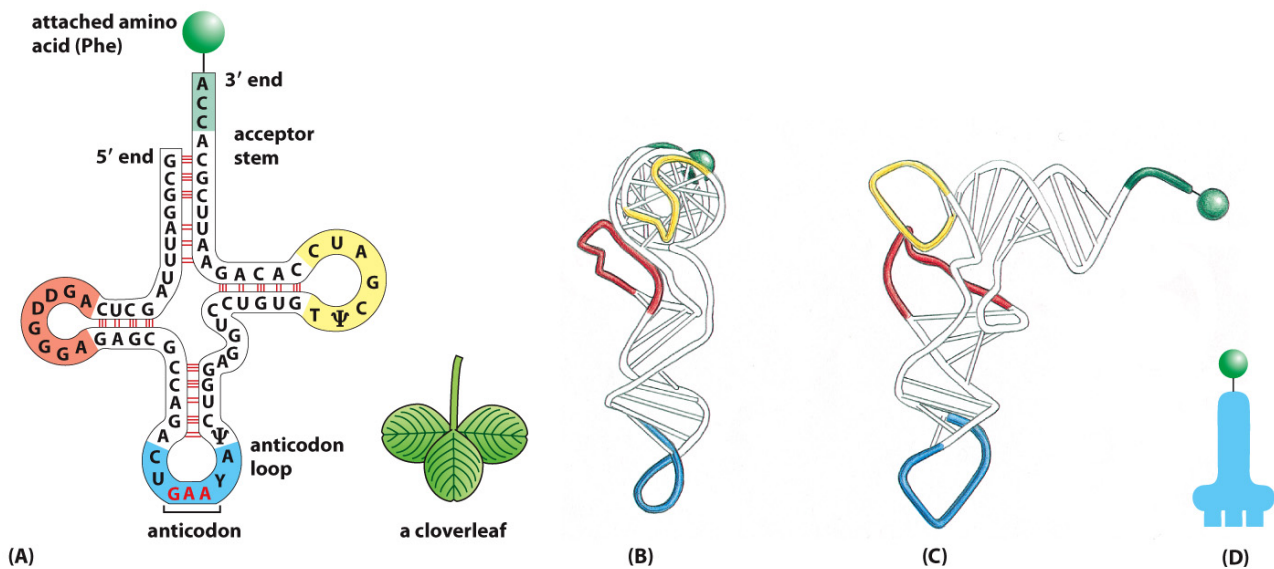
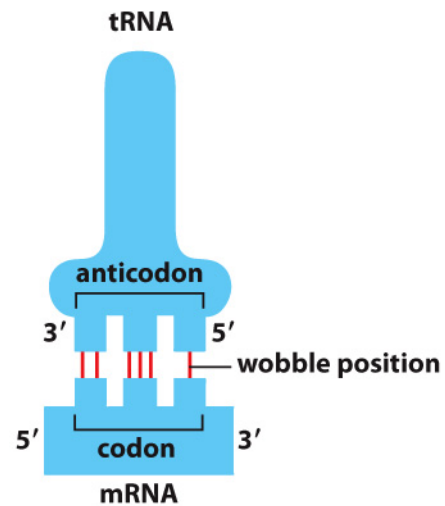


Figure 6-50 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 6–50 A tRNA molecule. A tRNA specific for the amino acid phenylalanine (Phe) is depicted in various ways. (A) The cloverleaf structure showing the complementary base-pairing (red lines) that creates the double-helical regions of the molecule. The anticodon is the sequence of three nucleotides that base-pairs with a codon in mRNA. The amino acid matching the codon/anticodon pair is attached at the 3' end of the tRNA. tRNAs contain some unusual bases, which are produced by chemical modification after the tRNA has been synthesized. For example, the bases denoted ψ (pseudouridine—see Figure 6–41) and D (dihydrouridine—see Figure 6–53) are derived from uracil. (B and C) Views of the L-shaped molecule, based on x-ray diffraction analysis. Although this diagram shows the tRNA for the amino acid phenylalanine, all other tRNAs have similar structures. (D) The tRNA icon we use in this book. (E) The linear nucleotide sequence of the molecule, color-coded to match (A), (B), and (C).

Figure 6–51 Wobble base-pairing between codons and anticodons. If the nucleotide listed in the first column is present at the third, or wobble, position of the codon, it can base-pair with any of the nucleotides listed in the second column. Thus, for example, when inosine (I) is present in the wobble position of the tRNA anticodon, the tRNA can recognize any one of three different codons in bacteria and either of two codons in eukaryotes. The inosine in tRNAs is formed from the deamination of adenosine (see Figure 6–53), a chemical modification that takes place after the tRNA has been synthesized. The nonstandard base pairs, including those made with inosine, are generally weaker than conventional base pairs. Codon–anticodon base-pairing is more stringent at positions 1 and 2 of the codon, where only conventional 3' base pairs are permitted. The differences in wobble base-pairing interactions between bacteria and eukaryotes presumably result from subtle structural differences between bacterial and eukaryotic ribosomes, the molecular 5' machines that perform protein synthesis. (Adapted from C. Guthrie and J. Abelson, in *The Molecular Biology of the Yeast *Saccharomyces*: Metabolism and Gene Expression*, pp. 487–528. Cold Spring Harbor, New York: Cold Spring Harbor Laboratory Press, 1982.)



bacteria

wobble codon base	possible anticodon bases
U	A, G, or I
C	G or I
A	U or I
G	C or U

eukaryotes

wobble codon base	possible anticodon bases
U	A, G, or I
C	G or I
A	U
G	C

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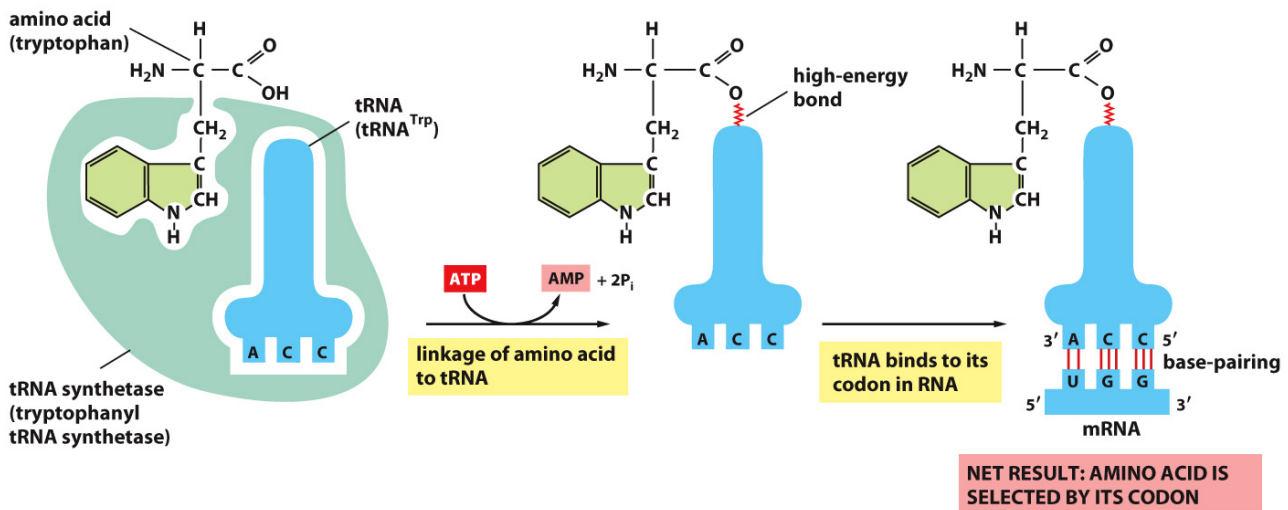


Figure 6-56 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 6–56 The genetic code is translated by means of two adaptors that act one after another. The first adaptor is the aminoacyl-tRNA synthetase, which couples a particular amino acid to its corresponding tRNA; the second adaptor is the tRNA molecule itself, whose anticodon forms base pairs with the appropriate codon on the mRNA. An error in either step would cause the wrong amino acid to be incorporated into a protein chain (Movie 6.6). In the sequence of events shown, the amino acid tryptophan (Trp) is selected by the codon UGG on the mRNA.

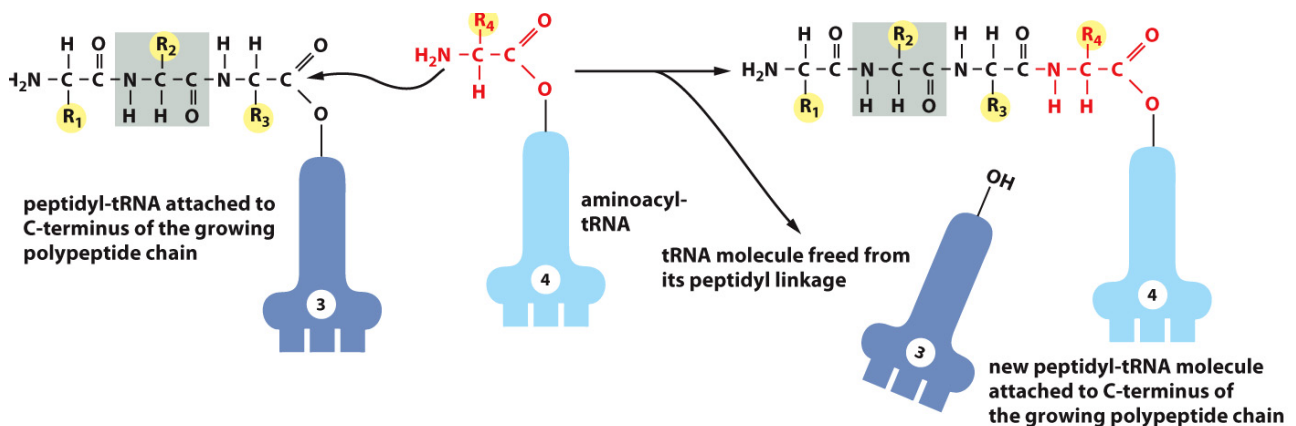


Figure 6-59 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 6–59 The incorporation of an amino acid into a protein. A polypeptide chain grows by the stepwise addition of amino acids to its C-terminal end. The formation of each peptide bond is energetically favorable because the growing C-terminus has been activated by the covalent attachment of a tRNA molecule. The peptidyl-tRNA linkage that activates the growing end is regenerated during each addition. The amino acid side chains have been abbreviated as R1, R2, R3, and R4; as a reference point, all of the atoms in the second amino acid in the polypeptide chain are shaded gray. The figure shows the addition of the fourth amino acid (red) to the growing chain.

Figure 6–64 Translating an mRNA molecule.

Each amino acid added to the growing end of a polypeptide chain is selected by complementary base-pairing between the anticodon on its attached tRNA molecule and the next codon on the mRNA chain. Because only one of the many types of tRNA molecules in a cell can base-pair with each codon, the codon determines the specific amino acid to be added to the growing polypeptide chain. The four-step cycle shown is repeated over and over during the synthesis of a protein. In step 1, an aminoacyl-tRNA molecule binds to a vacant A site on the ribosome. In step 2, a new peptide bond is formed. In step 3, the large subunit translocates relative to the small subunit, leaving the two tRNAs in hybrid sites: P on the large subunit and A on the small, for one; E on the large subunit and P on the small, for the other. In step 4, the small subunit translocates carrying its mRNA a distance of three nucleotides through the ribosome. This “resets” the ribosome with a fully empty A site, ready for the next aminoacyl-tRNA molecule to bind. As indicated, the mRNA is translated in the 5′-to-3′ direction, and the N-terminal end of a protein is made first, with each cycle adding one amino acid to the C-terminus of the polypeptide chain (Movie 6.7 and Movie 6.8).

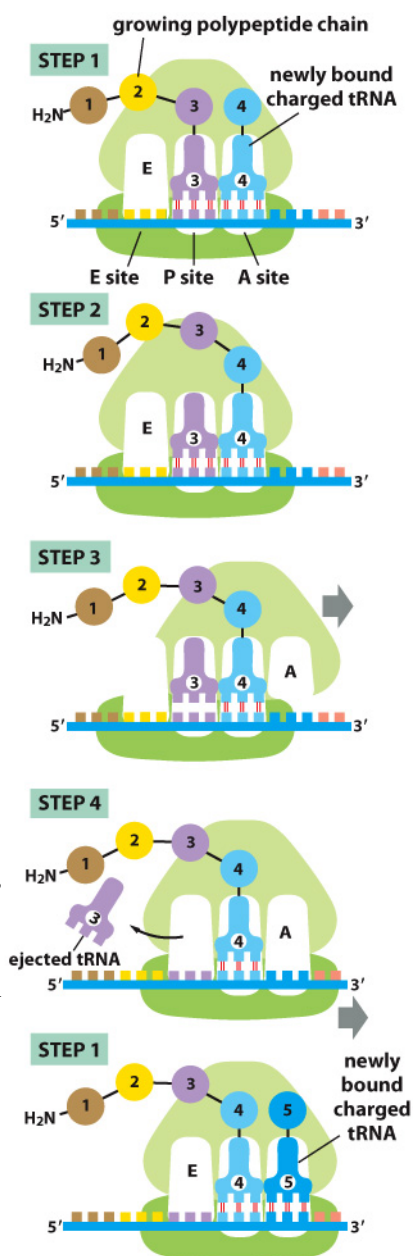


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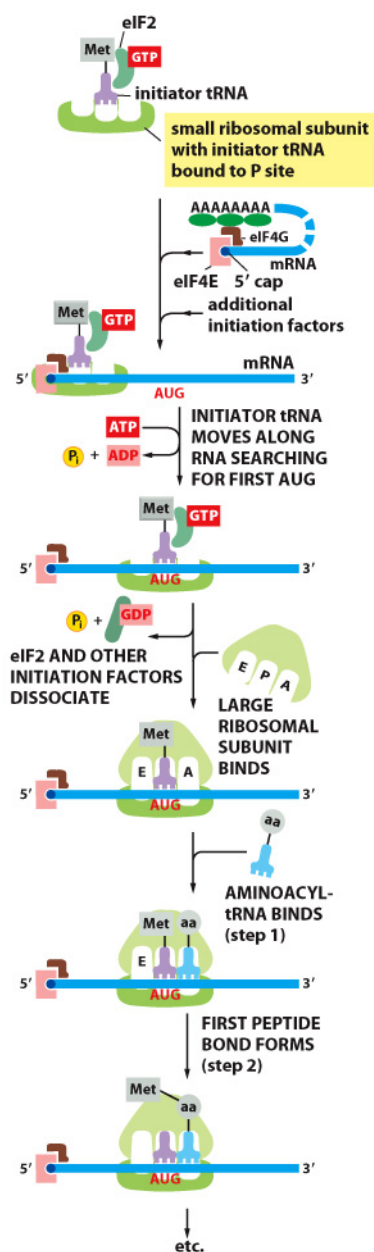


Figure 6–70 The initiation of protein synthesis in eukaryotes. Only three of the many translation initiation factors required for this process are shown. Efficient translation initiation also requires the poly-A tail of the mRNA bound by poly-A-binding proteins, which, in turn, interact with eIF4G (see Figure 6–38). In this way, the translation apparatus ascertains that both ends of the mRNA are intact before initiating protein synthesis. Although only one GTP-hydrolysis event is shown in the figure, a second is known to occur just before the large and small ribosomal subunits join. In the last two steps shown in the figure, the ribosome has begun the standard elongation cycle, depicted in Figure 6–64.

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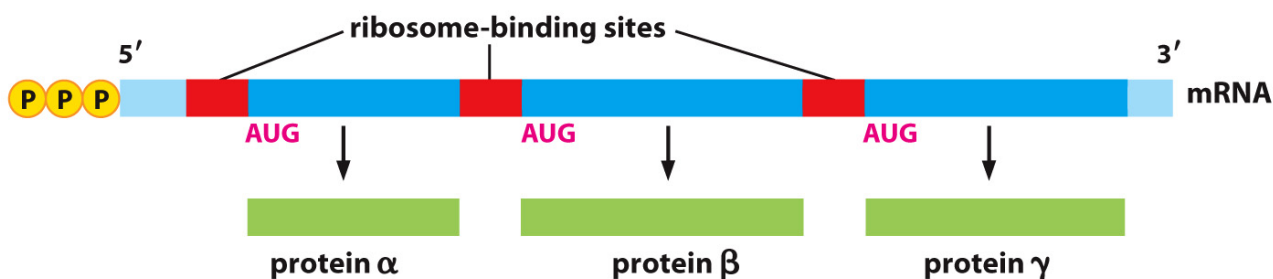


Figure 6-71 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 6–71 Structure of a typical bacterial mRNA molecule. Unlike eukaryotic ribosomes, which typically require a capped 5' end on the mRNA, prokaryotic ribosomes initiate translation at ribosome-binding sites (Shine–Dalgarno sequences), which can be located anywhere along an mRNA molecule. This property of their ribosomes permits bacteria to synthesize more than one type of protein from a single mRNA molecule.